

Microcosm Dynamics: an open, scored dynamic microsimulation model

Max Ghenis

September 2026

Abstract

Microcosm Dynamics is an open dynamic microsimulation model for the United States. It fits the transitions of individual lives on the Panel Study of Income Dynamics — earnings, marriage and divorce, births, disability, mortality and claiming — runs them forward year by year, and computes Social Security benefits from the resulting histories. Its gated components are scored on held-out or external data under terms locked before each scored run, any later change to those terms is a public amendment, and every scored run is committed whether it passes or fails. The earnings, family, disability and population-transport gates pass, as does a temporal holdout that fits the engine on data through 2014 and scores its projection against what the PSID recorded from 2015 to 2019. Four registered blind tests compare the model with DYNASIM3’s published results. A one-point cut to cost-of-living adjustments lands within 0.74 percentage points of DYNASIM3 in every age group. A full retirement age of 68 differs by 4.2 points at ages 65 to 69, a 13 percent benefit cut raises poverty at 67 by 0.47 points against DYNASIM3’s 2, and a minimum benefit reaches more women and fewer men than DYNASIM3 projects. The projection tests do not yet draw forward earnings, apply the retirement earnings test or model claiming responses, and benefits are computed with Python rules; the Axiom rules engine so far reproduces only their average indexed monthly earnings, computed with statutory computation years rather than the fixed 35 the projection tests count. Those are the next stages of work.

1 Introduction

Dynamic microsimulation ages a population of individual people through time: each year it draws their earnings, marriages, disabilities, deaths and program claims, then applies the rules of programs like Social Security to the lives that result. Analysts use it to answer questions a single-year snapshot cannot, such as who a benefit cut reaches decades from now, or how a change to the retirement age plays out across cohorts. The United States models that answer these questions — the Urban Institute’s DYNASIM, the Social Security Administration’s MINT and the Congressional Budget Office’s long-term model, CBOLT — run inside their institutions on code and administrative records that are not public, so outside analysts cannot rerun them or reproduce their results (Favreault et al. 2015; Smith et al. 2021; Congressional Budget Office 2018).

Microcosm Dynamics is an open dynamic microsimulation model. It extends Microcosm, PolicyEngine’s open cross-sectional microdata for the United States, with a longitudinal layer built from the Panel Study of Income Dynamics (PSID). Anyone can read its code, rerun its estimates and check its claims, under the MIT license at github.com/PolicyEngine/microcosm-dynamics.

Each gated module is scored by a gate: a test whose cells and tolerances are locked before the scored run, on held-out or external data. A gate’s terms change only by a public amendment made before the next scored run; the gates for generated earnings histories, family formation, population transport and the temporal holdout pass under amendments adopted after an earlier candidate failed (Section 4).

Earnings, family formation, disability and the transport onto the base population each pass their gates; claiming, disability-insurance entitlement and entrants run without one, and the projection gate certifies nothing about mortality drift. Every scored run, pass or fail, stays in the repository.

Four registered blind tests compare the model with DYNASIM3's published results. For each, the comparator values were transcribed and sealed before the run, the plan was posted publicly, and the registered run was scored once. A one-point cut to cost-of-living adjustments matches: the model's 2030 benefit changes fall within 0.74 percentage points of DYNASIM3's in every age group, 0.38 points on average. The other three tests each locate a specific place where the two models differ. Raising the full retirement age to 68, the model's benefit cut for people aged 65 to 69 in 2030 is 4.2 points shallower than DYNASIM3's. A 13 percent benefit cut raises adjusted poverty at age 67 in the 1936–45 birth cohort, which the model observes at odd birth years, by 0.47 points, against DYNASIM3's 2 points. A minimum benefit reaches 7.7 percent of beneficiaries aged 62 and older under its headline option, against DYNASIM3's 6.3 percent. A gap marks a disagreement between two models; neither side is ground truth, and for the last two tests the model's figures come from incomes already observed in the PSID while DYNASIM3's are projections published in 2004 and 2006.

The model does not yet project forward earnings into these comparisons. Its forward earnings law, fitted on PSID earnings through 2014 for ages 25 to 64, passes its own gate, but the DynaSim projections start in 2010 and need earnings for people in their sixties, where the retirement earnings test withholds benefits from beneficiaries who work before full retirement age. Connecting the two, adding the earnings test and behavioral claiming, and executing the benefit rules in the Axiom rules engine are the next stages of work.

2 The model

Microcosm Dynamics fits the transitions of individual lives on the Panel Study of Income Dynamics, then runs those transitions forward year by year on a population of people. Three layers do the work: a population to start from, a projection engine that moves each person through a year, and a rules layer that turns the resulting lives into Social Security benefits.

2.1 The starting population

Two kinds of starting population exist. The first is Microcosm's certified representative frame for the United States, built from the Current Population Survey: 166,302 people in 57,240 households with a 2024 reference period (`gates.yaml`, `gate_w1`). The transitions are estimated on the PSID, a panel that follows families for decades, and a transport gate tests whether those PSID-estimated processes reproduce the frame's cross-sectional joint distributions of earnings, age, sex, marital status and household composition when carried onto it.

The second kind is a closed PSID cohort, which a comparison builds for itself from label-verified PSID readers (`cohorts/`). The DynaSim tests use cohorts of this kind. The two projection tests, for example, start from 11,405 members of the 2011 PSID wave, whose earnings histories run from 1968 through 2010, taken from the PSID where it reports them and imputed or set to zero where it does not.

2.2 The projection engine

The engine advances a person-year slice one year at a time through eight steps in a fixed order: mortality, aging, the marital core, fertility, disability, earnings, claiming and household composition (`engine/loop.py`). Each step is an injected adapter that owns its own mathematics; the loop owns only the order and the flow of marital state, which fertility and household composition both read from the same step-3 result so that no later step can simulate a second, inconsistent marriage history.

People can also enter the population as scheduled immigrant cohorts, through a dedicated seam in the loop; births come from the fertility step.

Every random draw comes from a registered stream keyed by draw, period and module, and by person where needed (`engine/rng.py`), so a projection repeats exactly given its seed. The fitted components come from a refit bundle estimated only on data dated at or before a boundary year, 2014 for the current bundle (`engine/refit.py`). That boundary is what lets the engine be scored out of sample: it can be fitted on 2014 and earlier, projected across the PSID’s later waves, and compared with what the PSID actually recorded. The comparison is scored on 2015 to 2019; the 2020–2022 window, classed as an exogenous shock outside the model’s scope, is published as a diagnostic and scored in no gated cell (`gates.yaml`, `gate_m6`). The boundary constrains the fitted parameters, not every design choice: the gate records that its scored cells, and the structure of the forward earnings law, were selected with the later waves in view. The engine keeps the two uses apart (`engine/support.py`): a scored holdout run may condition on who the PSID observed in later waves, but an open forward projection cannot see the future and refuses that information.

2.3 The rules layer

The rules layer computes Social Security benefits from the lives the engine produces. A Python oracle (`ss/`) computes average indexed monthly earnings (AIME), the primary insurance amount (PIA), claiming-age adjustments, and spouse and survivor benefits, following 42 U.S.C. 415 and 402, with benefit parameters drawn from the PolicyEngine-US parameter tree. It can count either a fixed 35 computation years for every worker or the statutory number (`ss/statutory_aime.py`), which is smaller for workers who reach 62 with fewer elapsed years. The oracle is the reference the model’s rules engine is checked against. That rules engine is Axiom, which encodes the statute as machine-readable rules; a bridge (`axiom_benefit_bridge.py`) sends retained earnings histories through the actual Axiom engine. On the 7,486 members of the 2011-wave cohort who reach age 62 between 1979 and 2030, Axiom’s AIME equals the oracle’s, with statutory computation years, for every person. The projection tests instead count a fixed 35 years, which gives a different AIME for 251 of those people, all born before 1929. That agreement covers AIME alone, computed as for an old-age beneficiary with no period of disability, and rests on declarations both sides share: PSID labor income counts as covered earnings, and years before 1968, after 2010 or missing from the panel are sent as zeros or imputed amounts, 194,465 of the 320,375 earnings rows sent. Axiom’s encodings of the Social Security rules have not yet been accepted, so the DynaSim tests that compute benefits do so with Python rules.

3 What the model covers today

Each component either passes a registered gate, runs without one, or remains to be built. Table 1 lists them; the gate artifacts are committed under `runs/`, and each verdict below is read from the artifact itself.

Table 1: What each component of Microcosm Dynamics does, how it is tested, and whether the four DynaSim tests use it.

Component	What the model does	Test and status	Used in the DynaSim tests
Recorded earnings	PSID head and spouse labor income, 1968–2022, with unreported years imputed or set to zero	Data, with imputed gaps	Yes, in exercises 1, 3 and 4

Component	What the model does	Test and status	Used in the DynaSim tests
Generated earnings histories	A backcast process that generates earnings careers	gate 1: passes against held-out panel moments (runs/gate1_rank_knn_v5.json)	No
Forward earnings	Rank-chain law fitted on 2014 and earlier, ages 25–64	gate m6: passes, 4 of 5 seeds (runs/gate_m6_candidate3_v1.json)	No
Earnings test	SSA’s exempt amounts for 2000–2026 and the statutory formula that projects later years	Tested against SSA’s published amounts; not applied	No
Mortality	Death probabilities by age band and sex	Drift not certified by gate m6	Exercises 1 and 3 use SSA’s 2004 period life table; exercise 2 annuitizes wealth with NCHS 2000 tables
Disability	Work-limitation incidence, recovery, occupancy and exits near full retirement age	gate m4: passes, 5 of 5 seeds (runs/gate_m4_hazard_v1.json)	No
SSDI entitlement	Awards, recoveries and conversion at full retirement age, at rates through 2008	Not gated	Yes, in exercises 1 and 3
Claiming	Claim-age distribution by sex and entitlement year	Not gated	Yes, in exercises 1 and 3, with no response to reform
Marriage and fertility	Marriage, divorce, remarriage, widowhood and births	gate 2: passes, 4 of 5 seeds (runs/gate2_hazard_v16.json)	No
Household composition	Who lives with whom	gate 2b: passes, 4 of 5 seeds (runs/gate2b_hazard_v9.json)	No
Marriage by earnings	The joint of marital status and earnings	gate 2c: passes, 4 of 5 seeds (runs/gate2c_hazard_v2.json)	No
Entrants	Scheduled immigrant cohorts; births come from the fertility step	Report-only; not gated	No
Base frame	Transport of PSID processes onto the CPS frame	gate w1: passes (runs/gate_w1_candidate4_v1.json)	No
Benefit rules	AIME, PIA, auxiliaries in the Python oracle	Reference implementation	Yes, in exercises 1, 3 and 4
Axiom execution	Statutory rules in the Axiom engine	AIME matches the oracle for 7,486 of 7,486 people	No
SSI, income and poverty	Static SSI response and an adjusted poverty measure	Registered test, exercise 2	Yes, in exercise 2, which cuts reported benefits directly

Component	What the model does	Test and status	Used in the DynaSim tests
Wealth and pensions	PSID wealth annuitized into income	No projection	Exercise 2 only

Earnings. The model works with three kinds of earnings. The first is recorded: PSID head and spouse labor income for income years 1968 to 2022, annual through 1996 and every other year after (`data/family.py`). Years the panel does not report are imputed or set to zero, and those make up about half of the career years behind the projection tests’ benefits. The benefit calculations read these careers directly: through 2010 in the two projection tests, and over a longer span of the same panel, with later odd years taken from the next wave’s reports, in the minimum-benefit test. The second is generated: a backcast process that produces earnings careers, which gate 1 scores against held-out panel moments relative to the panel’s own noise floor. The DynaSim tests use recorded careers rather than generated ones. The third is projected: a forward earnings law that draws, every other year, whether each person works and where they land in the earnings distribution, then scales that rank to the national average wage (`engine/forward_earnings.py`). The law is fitted on earnings observed through 2014 for people aged 25 to 64. It has no fitted distribution above 64: older people are drawn from its oldest age band, and people without a fitted 2014 earnings state receive a fixed zero (`engine/earnings_domain.py`). Gate m6 fits the whole engine on 2014 and earlier, projects forward, and scores the projected earnings persistence, growth, mobility, low earnings and zero-earnings rate in 2016 and 2018 against what the PSID recorded. The law passes, and that pass is its only certification: gate 1 scores a different, backward-looking process. The law’s parameters use only data through 2014, but its structure was selected with the later waves in view, which the gate discloses. The two DynaSim projection tests nonetheless draw no earnings after 2010 (`cola_track_a/adapters.py`): their cohort opens in 2010, four years before the law’s fitted boundary, and the law has no fitted distribution past 64, while the retirement earnings test those tests point to acts on beneficiaries in their sixties. The parameters of that test exist — SSA’s exempt amounts from 2000 to 2026, checked against SSA’s published history, with the statutory formula that projects them — but no projection applies them yet.

Family. Three gates cover family life. Gate 2 scores marriage, divorce, remarriage, widowhood, the stock of people in each marital state, and births. Gate 2b scores household composition, and gate 2c scores the joint of marital status and earnings. All three pass at four of five seeds. The DynaSim projection tests use none of this machinery: their marital step marks a person widowed when a linked spouse dies in the simulation, and draws no marriages or divorces.

Disability, mortality and claiming. The disability module models self-reported work limitation: its incidence, recovery, occupancy and exits near full retirement age. Gate m4 scores it against held-out PSID employment-status records and against the age shape and exit composition, never the levels, of SSA’s 2023 disability-insurance statistics, and it passes at all five seeds. Work limitation is a different concept from a Social Security Disability Insurance award, so the projection tests use a separate, ungated SSDI entitlement component — awards, recoveries and conversion at full retirement age — at rates observed through 2008 (`engine/di_entitlement.py`). Mortality in the engine is a table of death probabilities by age band and sex; gate m6 certifies nothing about how mortality drifts over time, and the projection tests use SSA’s 2004 period life table scaled year by year to the 2008 Trustees Report. Claiming draws a claim age from a distribution by sex and entitlement year, which the projection tests cap at its 2008 row. No claim responds to a reform, which is why the retirement-age test keeps each person’s projected claim age under the reform and registers claiming responses only as alternatives.

Population. The transport gate, w1, passes: the PSID-estimated processes reproduce the CPS frame on its 44 gated cells: 43 cross-sectional joints and one ordering check. New people can enter a projection through the entrant schedule, but that schedule is report-only; no gate scores it, and nothing it produces enters a fitted law.

Rules. The two projection tests and the minimum-benefit test compute benefits with Python rules; the projection tests count a fixed 35 computation years for every worker (`cola_track_a/benefits.py`). Exercise 2 applies no benefit rules: it cuts reported Social Security income by 13 percent in a Python income concept, then applies a static SSI offset and computes an adjusted poverty measure, annuitizing PSID wealth into income. Axiom reproduces the oracle’s AIME for all 7,486 cohort members tested, which establishes that the two compute the same number from the same records, not that either number is a person’s true AIME.

4 How the model is scored

A component enters the model when it passes a gate. Each gate names what it covers, the data it holds out, the cells it scores and the tolerance for each cell, and all of that is written into `gates.yaml` and locked before a candidate’s scored run. A candidate is registered, run once against the gate, and committed under `runs/` whatever the verdict; a failed candidate stays in the record beside the one that passes. A gate’s terms change only by a public amendment made before the next scored run, and four of the passing verdicts rest on amendments adopted after an earlier candidate failed. Gate 1 passes a candidate identical to its failed predecessor under two amendments: one moved a failing distinguishability test to report-only and added a gated benefit-space block, and the other scored another such test by its mean across twenty seeds (`gates.yaml`, `gate_1`; `runs/gate1_rank_knn_v5.json`). Gate 2 passes its sixteenth candidate, a changed model, under an estimator amendment and a tranche split adopted after earlier candidates failed (`gates.yaml`, `gate_2`). Gate w1 passes a candidate identical to its failed predecessor under three amendments, each adopted after a failed candidate, which together made 21 cells report-only and narrowed its ordering check (`runs/NOTES-gate_w1_candidate4.md`). Gate m6 passes its third candidate after an amendment restricted its earnings cells to people with a fitted 2014 state and re-derived its floors; its first candidate failed before that amendment and its second after it (`docs/amendments/gate_m6_amendment_1_closed_domain_floors.md`).

Four kinds of gate exist. Holdout gates score a model on people it was not fitted on: the family gates hold out 20 percent of persons, on the same five seeds for every method compared (`gates.yaml`, `holdout_protocol`). Each cell’s tolerance is set relative to a noise floor, the gap between two disjoint halves of the same real panel, so the gate judges a model against the sampling variation of the data it is scored on (`gates.yaml`, `noise_floor`). A seed passes when every gated cell holds, and the family gates pass when at least four of five seeds do. Anchor gates score a module on a person-disjoint half of the panel and against external administrative publications; the disability gate compares the age shape and exit composition of its work-limitation estimates with SSA’s 2023 disability-insurance statistics, and gates no SSA level. The transport gate, w1, scores whether PSID-estimated processes reproduce the CPS frame’s cross-sectional joints. The temporal holdout, gate m6, fits the whole engine on data from 2014 and earlier, projects forward, and scores projected marital and disability flows for 2015 to 2019 and earnings moments for 2016 and 2018 against what the PSID recorded; the 2020–2022 window is published as a diagnostic. A gate also names what it does not certify: gate m6 certifies nothing about mortality drift, widowhood drift or entrants, and cells marked report-only are published without a tolerance.

The DynaSim comparisons (Section 5) follow the same discipline without a tolerance. DYNASIM3 is a published model, not ground truth, so each comparison reports its gaps and grades nothing.

The design bounds the claims the model makes by horizon. Over the long horizon, assumptions dominate the machinery: the 2025 Trustees Report puts the 75-year actuarial balance at -3.82 percent of taxable payroll under its intermediate assumptions, but anywhere from -0.33 percent under its low-cost assumptions to -8.60 percent under its high-cost ones, a spread more than twice the intermediate deficit itself (Board of Trustees, Federal Old-Age and Survivors Insurance and Federal Disability Insurance Trust Funds 2025). The design therefore sorts outputs into three tiers. The

first is distributional reform analysis under a fixed population and assumption path, where much of the demographic uncertainty is common to baseline and reform and cancels in the difference; the parts that do not cancel, such as claiming responses, are labelled, and claiming ranges anchor to the quasi-experimental record (Mastrobuoni 2009; Behaghel and Blau 2012). The second is near-term components — beneficiary counts, average benefits, covered earnings, claiming ages, disability incidence — that resolve against administrative publications each year, which gate 3 is designed to score. The third is the long horizon, which the design publishes as sensitivity surfaces over documented assumption ranges and computes on named assumption paths, including the Trustees’ intermediate path, rather than as a forecast of its own. Gate 3 is defined in `gates.yaml` and has not yet been run.

5 Four blind tests against DYNASIM3

Four registered tests compare Microcosm Dynamics with results the Urban Institute published from DYNASIM3. Two are 2030 projections of benefit reforms: a one-point cut to cost-of-living adjustments and a rise in the full retirement age to 68 (Urban Institute 2010). Two are static calculations on incomes already observed in the PSID: a uniform 13 percent benefit cut (Butrica and Uccello 2004) and a minimum benefit (Favreault et al. 2006). The COLA test matches. The other three each locate a specific place where the two models differ.

Every cell, registered alternative and timeline below is also published as an interactive scorecard at <https://microcosm.institute/dynamics/scorecard>, and each registration and result is a public comment on issue #42 at <https://github.com/PolicyEngine/microcosm-dynamics/issues/42>.

5.1 The protocol

Each test follows the same order. A separate comparator lane transcribes or digitizes DYNASIM3’s published values twice, independently, reconciles the two readings and seals the files under SHA-256 hashes. The model’s specification is ratified and frozen at a named commit. A forecaster who has not seen the sealed values predicts both the model’s result and the gap, and the forecast is hashed. The registration — specification hash, commit, primary row, registered alternatives and forecast — is posted publicly on issue #42 of the repository. The model then runs, and that run’s output is the only one scored. Its artifact is checked and committed before the comparison lane opens the seal, and the comparison is published whatever it shows.

No test has an acceptance threshold. Each result is reported, not graded, and no registered alternative can replace the primary row after the fact. The hashes establish that the comparator and artifact files did not change; the commit times and the recorded order of steps establish that the output was committed before the seal was opened. Neither establishes that no participant ever saw a comparator value, so each registration discloses its participants’ exposures, and a claim of blindness rests on those disclosures.

Table 2: The order of each test’s registration, run, commit and comparison.

Test	Registration posted (UTC)	Run	Artifact committed	Result posted
1. COLA cut	2026-09-23 23:06:06	23:06–23:31	23:40:38	23:42:53
3. Full retirement age 68	2026-09-25 22:35:30	22:36–23:07	23:27:01	23:40:59
2. 13 percent cut	2026-09-27 05:01:06	05:01:49–05:03:29	05:06:13	05:29:39

Test	Registration posted (UTC)	Run	Artifact committed	Result posted
4. Minimum benefit	2026-09-27 12:18:29	12:19:17–12:19:48	12:36:32	12:58:12

5.2 The projections

The two projection tests share one engine configuration, Track A. It ages a closed cohort seeded from the 2011 PSID wave to 2030 over 20 random draws, applying 2008-vintage mortality and disability rates and the statutory COLA and eligibility rules. Benefits are computed with the Python oracle, counting a fixed 35 computation years for every worker. Track A draws no earnings after 2010. Benefits rest on careers through 2010, on observed 2010 benefit amounts for people already retired at the opening, and on an approximation for disabled workers. The statistic is the percentage change in average Social Security benefits in 2030, a weighted ratio of scenario means, in five age groups.

Uncertainty on the model’s side has two measures. The draw standard deviation captures variation across the 20 projections. The half-sample floor is the mean absolute difference between two halves of the sample that share no person or family unit, averaged over five random splits; it measures how far the estimate moves on sampling alone. DYNASIM3’s values come from a figure, so each carries a digitization interval rather than a standard error.

5.2.1 Exercise 1: a one-point COLA cut

The reform reduces every annual cost-of-living adjustment by one percentage point from 2010. The comparator is Figure 2 of *Details Matter*, DYNASIM3 run 614 on 2008 Trustees assumptions (Urban Institute 2010).

Table 3: Exercise 1: percent change in average benefits in 2030 under a one-point COLA cut.

Age in 2030	Microcosm Dynamics	DYNASIM3	Gap (pp)	Half-sample floor
50–61	−7.77	−7.15	−0.62	0.27
62–64	−3.64	−3.79	+0.14	0.35
65–69	−6.26	−6.36	+0.10	0.12
70–79	−12.22	−11.94	−0.28	0.16
80+	−17.92	−18.66	+0.74	0.09

Every age group lands within 0.74 percentage points of DYNASIM3, with a mean absolute gap of 0.38 and the same ordering across all five groups. Two groups sit within the half-sample floor; three do not. The largest gap is at 80 and over, where DYNASIM3’s −18.66 percent lies beyond the deepest cut the model’s timing convention can produce for anyone in that group, −18.56 percent after 21 reduced increases. A registered alternative that counts the December 2030 increase brings 80+ within 0.06 points but makes every other group’s cut 0.8 to 1.5 points deeper than DYNASIM3’s, so no single registered timing convention matches all five groups. Different timing for the oldest cohorts, or a different mix of survivors on older benefit clocks, could explain it; both are untested hypotheses, and the published figure cannot tell them apart. At 50–61, a group almost entirely of disabled workers, the model’s cut is 0.62 points deeper; the durations of the opening disability stock and the 2008-vintage award and termination rates are the untested candidates.

Once the COLA clock is fixed, the reform acts mechanically on benefits already in payment, which makes this a weak test of behavioral dynamics. Six registered alternatives change one convention each; the entitlement clock widens the mean gap to 1.47 points, and the 2009 PSID wave leaves it at 0.37.

5.2.2 Exercise 3: a full retirement age of 68

The reform raises the full retirement age gradually from 2010, reaching 68 for people turning 62 in 2022 and later. The comparator is the same figure’s “Increase FRA to 68” bars (Urban Institute 2010). The test reuses exercise 1’s projection path for path and holds claiming ages fixed at their baseline values.

Table 4: Exercise 3: percent change in average benefits in 2030 when the full retirement age rises to 68.

Age in 2030	Microcosm Dynamics	DYNASIM3	Gap (pp)	Half-sample floor
50–61	–0.00	0.00 (reading)	–0.00	0.00
62–64	–4.73	–5.12	+0.39	0.09
65–69	–5.68	–9.90	+4.22	0.11
70–79	–5.24	–4.15	–1.09	0.07
80+	–0.75	–0.60	–0.14	0.11

The mean absolute gap is 1.17 points, and one group carries 72 percent of it: at 65 to 69, DYNASIM3’s cut is 1.74 times the model’s, a gap of 4.22 points. Nine of ten pairwise orderings agree. The 50–61 comparator bar has no visible height, so its zero is a digitizer’s reading bounded by the drawing’s resolution of about 0.09 points; the model’s estimate is consistent with it at that resolution.

Everyone aged 65 to 69 in 2030 was born between 1961 and 1965 and faces a 12-month rise in the full retirement age. At fixed claiming ages, that rise cuts a retired worker’s benefit by 6.25 to 7.69 percent and a spouse’s excess by at most 10 percent, and leaves a disabled worker’s benefit unchanged. Reaching DYNASIM3’s 9.90 percent through those factors would take a group made almost entirely of spouses’ excesses starting at 65, while disabled and converted-disabled benefits make up 17.8 percent of the model’s baseline dollars in that group. DYNASIM3’s bar therefore carries something beyond the age-factor reduction at fixed claim ages. Three untested hypotheses could account for it. A retirement earnings test extended to 68 would withhold benefits from people who keep working, and the model pays gross benefits and draws no earnings after 2010, so testing it needs forward earnings. A claiming response would move claims later. An average taken over everyone alive, rather than over recipients, would count a postponed claimant as a zero; a diagnostic on the model’s own claiming alternative moves the 65–69 change to –8.68 percent under that averaging. None of the eight registered alternatives, which vary the phase-in schedule, claiming response, statistic, benefit types, survivor rules and PSID wave, comes within 3.9 points at 65 to 69.

The first registration of this test, Registration 14, stopped at the runner’s membership guard after the primary row had been computed in memory, but before any value was printed or recorded, before any artifact existed and before the seal was opened. The scored run is Registration 15, on the reviewed fix, with the forecast unchanged.

5.3 The static calculations

The other two tests apply each reform to incomes already observed in the PSID and count who is affected. They run no projection and model no behavioral response. The model’s figures describe realized incomes; DYNASIM3’s are projections published in 2004 and 2006. A gap in either test can therefore reflect the vintage of the population as well as the two models’ rules.

5.3.1 Exercise 2: a uniform 13 percent benefit cut

The reform cuts every Social Security benefit by 13 percent from 2004. The comparator is Tables 19 and 21 of *How Will Boomers Fare at Retirement?*, which report adjusted poverty at age 67 for the 1936–45 birth cohort with and without the cut (Butrica and Uccello 2004). The statistic is the change in the poverty rate, in percentage points. The model observes 483 people at exactly age 67, born in 1937, 1939, 1941, 1943 and 1945, and attaches each to their family’s income and poverty threshold. The printed levels are whole percentages, so each printed change carries an open rounding interval of one point either side.

Table 5: Exercise 2: change in adjusted poverty at 67 under a 13 percent benefit cut.

Group	n	Microcosm Dynamics	DYNASIM3	Gap (pp)
All	483	+0.47	+2	−1.54
Women	249	+0.35	+3	−2.65
Men	234	+0.60	+1	−0.40
Married	331	+0.60	+1	−0.40
Widowed	44	+0.45	+2	−1.55
Divorced	60	0	+5	−5.00
Never married	14	0	+5	−5.00

The cut raises the model’s poverty rate by 0.47 points, against DYNASIM3’s 2, and the model’s change is smaller in all seven groups. Men and married people fall inside DYNASIM3’s rounding interval; the other five groups do not. Only four people in the sample cross the poverty line, and each average-weight observation is worth 0.21 points of the headline. Divorced and never-married people show no change because nobody in those groups crosses the line, and a cell in which nobody crosses the line has no estimable uncertainty, whatever its size. The model’s poverty levels are also lower than DYNASIM3’s, in six of seven groups before the cut and in all seven after it.

Ten registered alternatives change one input each: the birth years covered, the SSI response, the income unit, asset income, employer pension balances, the poverty standard, the life table and the threshold schedule. Five of them leave every poverty outcome unchanged, and none brings the headline into DYNASIM3’s interval; covering all ten birth years comes closest, at +0.62. The untested hypotheses include PSID’s thinner coverage of low-income unmarried people, whole-family income standing in for an individual’s resources, income sources the report does not name, self-reported benefits that may be net of Medicare premiums, and the difference between incomes realized in 2004–2012 and a projection built on 2002 Trustees assumptions.

5.3.2 Exercise 4: a minimum benefit

The reform sets a minimum benefit tied to years of work, in four versions that pair price or wage indexing with a standard or generous schedule, each financed by an across-the-board cut. The comparator is Table 6 of *Minimum Benefits in Social Security*, the 2025 shares of beneficiaries aged 62 and older who receive the minimum, for everyone, men and women (Favreault et al. 2006). The model applies each version to the 2,135 Social Security recipients observed in the PSID’s 2023 wave, with income year 2022, and counts those whose benefit rests on the minimum through their own record or a linked spouse’s or survivor’s benefit.

Table 6: Exercise 4: percent of beneficiaries aged 62 and older receiving the minimum benefit.

Option	Group	Microcosm Dynamics	DYNASIM3	Gap (pp)
2 (headline)	All	7.75	6.3	+1.45
2	Men	2.36	4.8	−2.44
2	Women	11.67	7.6	+4.07
3	All	11.12	11.3	−0.18
3	Men	3.23	8.7	−5.47
3	Women	16.86	13.4	+3.46
4	All	24.56	18.7	+5.86
4	Men	13.42	14.4	−0.98
4	Women	32.66	22.0	+10.66
5	All	31.47	29.3	+2.17
5	Men	19.24	24.4	−5.16
5	Women	40.36	33.2	+7.16

Both models rank the four options in the same order and put women above men under every option. The model’s shares are lower than DYNASIM3’s for men and higher for women in every option, with a mean absolute gap of 4.09 points across the twelve cells. Aggregation hides the split: option 3 lands within 0.18 points for everyone while differing by 5.47 points for men and 3.46 for women, in opposite directions. The untested hypotheses include the difference between a realized 2022 population and a 2025 projection, the source of each person’s primary insurance amount and the completeness of PSID earnings histories, disability-origin records, and 60 auxiliary beneficiaries the model cannot link to a worker. A registered alternative that takes each worker’s PIA from the benefit they receive, instead of from their earnings history, cuts the mean gap to 1.60 points; it remains an alternative and cannot replace the primary row.

5.4 What the four tests show and what they cannot

The four tests show a public comparison procedure that preserves agreement and disagreement alike. The COLA profile agrees to within 0.74 points everywhere. The full retirement age differs by 4.22 points in one age group. The poverty response is smaller across the board. Minimum-benefit receipt splits by sex. The blind forecasts, scored separately, placed every projection estimate inside its interval but left one gap in each projection test outside its interval, and in the two static tests put the model’s own values outside their intervals more often than DYNASIM3’s.

They cannot establish parity with DYNASIM4, which none of them tests, or certify the model. Three of them compute benefits with Python rules and exercise 2 cuts reported benefits directly; executing the rules in the Axiom rules engine is separate work. The two static tests do not exercise the projection engine at all, and the two projection tests exercise it without forward earnings, an earnings test or behavioral claiming. A gap is a disagreement between two models. In exercises 1 and 3 neither side can be checked against an observed outcome, because neither reform is current law; in exercises 2 and 4 the model’s figures come from observed incomes and DYNASIM3’s from projections. The registered alternatives measure how sensitive each result is to the model’s own conventions. They do not reveal how DYNASIM3 computes its results, and every candidate explanation above remains a hypothesis until a registered test addresses it.

6 What remains to build

The largest gap between the model’s components and its DynaSim projections is earnings. The forward earnings law passes its gate, but the projections that DYNASIM3’s reform results require start in 2010,

run to 2030, and turn on people in their sixties, where the law does not yet reach. The planned work closes that gap first and builds the benefit mechanics that depend on it second; each step below is a design, and each enters the model only through its own registered gate.

Forward earnings in the projection. The plan builds a bridge that carries a 2010 cohort’s earnings state to the law’s 2014 boundary, scored by a new transport gate on the 2012 and 2014 waves, whose registration must disclose that those waves already informed the model’s selection. A sibling fitter extends the law to ages 25 through 69, with a separate gate for each of the bands 62–64, 65–66 and 67–69. A backtest from a 2006 boundary across the 2006–2022 waves provides the evidence any 2030 claim needs but cannot supply on its own, because those waves already informed the model’s choices. A history recorder then keeps labor earnings, covered earnings and earnings countable under the retirement earnings test as separate series from 2010 to 2030, with a monthly calendar underneath.

The retirement earnings test. The test’s parameters exist and match SSA’s published history; applying them is next. The plan adds an annual ledger for entitlement, provisional suspension, final deduction and the grace year, checked against SSA’s worked examples, and separate modules for the adjustment that restores withheld months at full retirement age and for benefit recomputation on later earnings. Integrated with fixed claiming, this machinery makes the hypothesis behind the retirement-age test’s largest gap — an earnings test extended to 68 — testable for the first time.

Gross benefits and dual entitlement. A gross-benefit layer adds the family maximum, age and dual-entitlement reductions and the monthly family payment, checked against statutory worked examples and, where a pinned rule exists, against Axiom. A separate registered rerun of the two projection tests applies statutory computation years to disability benefits and adds dual entitlement to disability and spouse benefits. Its code, in an open pull request, is tested on invented data; the rerun awaits its own registration and is labelled post hoc because it follows the published results.

Claiming. The engine draws claim ages from a fixed distribution. The plan replaces it with a claiming hazard estimated on insured people who have not yet claimed, month by month, and — only if data access allows a credible design — a claiming response identified from policy variation. A response to a full retirement age of 68 would still be an extrapolation beyond any observed policy.

Family and population in the projection. The family models pass their gates but do not yet run inside a DynaSim projection. The plan composes the gated marriage and divorce models with widowhood triggered by a spouse’s death, redraws initial states, declares which populations the projection covers, and scores the result with a registered projection gate and a joint validation of earnings, claiming, disability, payments and survivor pathways. Entrants — births and immigrants — remain report-only until an admitted initialization supports them.

Axiom execution. Axiom reproduces the oracle’s AIME for every person tested. Running the full benefit rules in Axiom requires its encodings of those rules to be accepted and then carried from AIME through the PIA, auxiliaries and the earnings test.

Wealth, pensions and the long horizon. The model reads PSID wealth and annuitizes it for income measures but does not project it. Gate m6 certifies nothing about mortality drift, which bounds any projection far beyond the 2030 horizon of the DynaSim tests.

Held-out targets. Two new blind targets follow the same protocol as the four tests. The first is the 1946–55 birth cohort from the same 2004 report as exercise 2, whose DYNASIM3 values sit under the existing seal and stay out of the builders’ reach; its specification extends the PSID income and wealth mappings to the 2015–2023 waves. The second, conditional on registered conventions for an operational formula the source report leaves unstated, is the progressive price indexing option from the same 2010 figure as exercises 1 and 3. The first target that needs a projected population follows once the projection gates pass.

7 Related work

Dynamic microsimulation has a long literature (Harding 1990; Li and O’Donoghue 2013), and a small set of models carries the United States Social Security debate. The Urban Institute’s DYNASIM projects the incomes and retirement outcomes of the population (Favreault and Smith 2004; Favreault et al. 2015), with DYNASIM4 its current generation (Cosic et al. 2024). The Social Security Administration’s MINT projects the income of future retirees (Smith et al. 2021; Social Security Administration 2024), and the Congressional Budget Office’s CBOLT produces its long-term projections (O’Harra et al. 2004; Congressional Budget Office 2018, 2024), including a microsimulation of individual earnings (Schwabish and Topoleski 2013). Other United States models also project Social Security outcomes (Penn Wharton Budget Model 2025; Chanwong 2026). Outside researchers cannot rerun DYNASIM, MINT or CBOLT on their own, because their code or the administrative records they rely on are not public; the Cato model publishes its code and draws on public data (Chanwong 2026).

Open tools for dynamic microsimulation exist outside the United States Social Security domain: SimPaths models the life course (Bronka et al. 2025), LIAM2 (Menten et al. 2014) and OpenM++ (OpenM++ development team 2026) are platforms for building such models, and Destinie 2 (Blanchet et al. 2010) and microWELT (Spielauer et al. 2020) model pensions and transfers in ageing European populations. Work on weighting dynamic models (Dekkers and Cumpston 2012) and on aligning simulated outcomes with external totals (Li and O’Donoghue 2014) addresses problems any model of this kind faces. Microcosm Dynamics extends PolicyEngine’s open cross-sectional microsimulation (PolicyEngine 2026; Ghenis and Woodruff 2024) with a longitudinal layer, and it differs from the models above in one respect the four DynaSim tests make concrete: every component, test and comparison is public, including the comparisons where it disagrees with DYNASIM3.

References

- Behaghel, Luc, and David M Blau. 2012. “Framing Social Security Reform: Behavioral Responses to Changes in the Full Retirement Age.” *American Economic Journal: Economic Policy* 4 (4): 41–67.
- Blanchet, Didier, Sophie Buffeteau, Emmanuelle Crenner, and Sylvie Le Minez. 2010. *The New Destinie 2 Microsimulation Model: Main Characteristics and Illustrative Results*. Document de Travail Nos. G2010-13. INSEE.
- Board of Trustees, Federal Old-Age and Survivors Insurance and Federal Disability Insurance Trust Funds. 2025. *The 2025 Annual Report of the Board of Trustees of the Federal Old-Age and Survivors Insurance and Federal Disability Insurance Trust Funds*. Social Security Administration. <https://www.ssa.gov/oact/TR/2025/tr2025.pdf>.
- Bronka, Patryk, Justin van de Ven, Daniel Kopasker, S. Vittal Katikireddi, and Matteo Richiardi. 2025. “SimPaths: An Open-Source Microsimulation Model for Life Course Analysis.” *International Journal of Microsimulation* 18 (1): 95–133. <https://doi.org/10.34196/ijm.00318>.
- Butrica, Barbara, and Cori E. Uccello. 2004. *How Will Boomers Fare at Retirement?* Nos. 2004-05. AARP Public Policy Institute.
- Chanwong, Krit. 2026. *Social Security Cato Model*. https://github.com/kchanwong/social_security_cato_model.
- Congressional Budget Office. 2018. *An Overview of CBOLT: The Congressional Budget Office Long-Term Model*. Congressional Budget Office. <https://www.cbo.gov/publication/53667>.

- Congressional Budget Office. 2024. *CBO’s 2024 Long-Term Projections for Social Security*. Congressional Budget Office. <https://www.cbo.gov/publication/60392>.
- Cosic, Damir, Richard W. Johnson, and Karen E. Smith. 2024. *Urban’s Dynamic Simulation of Income Model 4*. Fact sheet. Urban Institute. <https://www.urban.org/research/publication/urbans-dynamic-simulation-income-model-4>.
- Dekkers, Gijs, and Richard Cumpston. 2012. “On Weights in Dynamic-Ageing Microsimulation Models.” *International Journal of Microsimulation* 5 (2): 59–65.
- Favreault, Melissa M., Gordon B. T. Mermin, and C. Eugene Steuerle. 2006. *Minimum Benefits in Social Security*. Urban Institute.
- Favreault, Melissa M., Karen E. Smith, and Richard W. Johnson. 2015. *The Dynamic Simulation of Income Model (DYNASIM): An Overview*. Research Report. Urban Institute. <https://www.urban.org/research/publication/dynamic-simulation-income-model-dynasim-overview>.
- Favreault, Melissa, and Karen E. Smith. 2004. *A Primer on the Dynamic Simulation of Income Model (DYNASIM3)*. The Urban Institute. <https://www.urban.org/sites/default/files/publication/71226/410961-A-Primer-on-the-Dynamic-Simulation-of-Income-Model-DYNASIM-.PDF>.
- Ghenis, Max, and Nikhil Woodruff. 2024. *Beta-Launch: PolicyEngine’s Enhanced Microdata for Policy Analysis*. PolicyEngine research post. <https://policyengine.org/us/research/enhanced-cps-beta>.
- Harding, Ann. 1990. *Dynamic Microsimulation Models: Problems and Prospects*. Welfare State Programme Discussion Paper No. 48. Suntory-Toyota International Centre for Economics; Related Disciplines (STICERD), London School of Economics.
- Li, Jinjing, and Cathal O’Donoghue. 2013. “A Survey of Dynamic Microsimulation Models: Uses, Model Structure and Methodology.” *International Journal of Microsimulation* 6 (2): 3–55. <https://doi.org/10.34196/ijm.00082>.
- Li, Jinjing, and Cathal O’Donoghue. 2014. “Evaluating Binary Alignment Methods in Microsimulation Models.” *Journal of Artificial Societies and Social Simulation* 17 (1): 15. <https://doi.org/10.18564/jasss.2334>.
- Mastrobuoni, Giovanni. 2009. “Labor Supply Effects of the Recent Social Security Benefit Cuts: Empirical Estimates Using Cohort Discontinuities.” *Journal of Public Economics* 93 (11-12): 1224–33.
- Menten, Gaëtan de, Gijs Dekkers, Geert Bryon, Philippe Liégeois, and Cathal O’Donoghue. 2014. “LIAM2: A New Open Source Development Tool for Discrete-Time Dynamic Microsimulation Models.” *Journal of Artificial Societies and Social Simulation* 17 (3): 9. <https://doi.org/10.18564/jasss.2574>.
- O’Harra, Josh, John Sabelhaus, and Michael Simpson. 2004. *Overview of the Congressional Budget Office Long-Term (CBOLT) Policy Simulation Model*. Technical Paper Nos. 2004-01. Congressional Budget Office. <https://www.cbo.gov/publication/15188>.
- OpenM++ development team. 2026. *OpenM++: Open Source Microsimulation Platform*. <https://openmpp.org>.

- Penn Wharton Budget Model. 2025. *Penn Wharton Budget Model: Microsimulation*. <https://budgetmodel.wharton.upenn.edu/model/microsimulation/>.
- PolicyEngine. 2026. *PolicyEngine: Open-Source Tax-Benefit Microsimulation*. <https://policyengine.org>.
- Schwabish, Jonathan A., and Julie H. Topoleski. 2013. *Modeling Individual Earnings in CBO's Long-Term Microsimulation Model*. Working Paper Nos. 2013-04. Congressional Budget Office. <https://www.cbo.gov/publication/44306>.
- Smith, Karen E., Aaron R. Williams, and Stipica Mudrazija. 2021. *Modeling Income in the Near Term: Version 8*. Urban Institute. <https://www.urban.org/research/publication/modeling-income-near-term>.
- Social Security Administration. 2024. *Projection Methodology: Modeling Income in the Near Term, Version 8 (MINT8)*. <https://www.ssa.gov/policy/docs/projections/methodology.html>.
- Spielauer, Martin, Thomas Horvath, and Marian Fink. 2020. *microWELT: A Dynamic Microsimulation Model for the Study of Welfare Transfer Flows in Ageing Societies from a Comparative Welfare State Perspective*. WIFO Working Papers 609/2020. Austrian Institute of Economic Research (WIFO).
- Urban Institute. 2010. *Distributional Effects of Alternative Social Security Reforms: Details Matter*. Fact Sheet on Retirement Policy. Urban Institute.